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Palaeomagnetic and K–Ar age studies of a 6 km-thick Cretaceous section from the Chilean Andes

H. C. Palmer and A. Hayatsu *Department of Geophysics, The University of Western Ontario, London, Ontario, Canada N6A 5B7*

W. D. MacDonald *Department of Geological Sciences, State University of New York, Binghampton, New York 13901, USA*

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Summary. Thirty-six palaeomagnetic sampling sites distributed within 6000 m of dominantly andesitic flows and tuffs of Cretaceous age from the La Serena area, Chile confirm the normal polarity bias of the Cretaceous period. Af, thermal and limited chemical demagnetization techniques have been used in testing the stability of the remanent magnetization isolated in samples from these sites. A positive fold test in the Quebrada Marquesa Formation, the second lowest in the stratigraphic pile, confirms that the magnetization isolated is pre-Tertiary in age. Ages calculated by the K–Ar whole rock method however, appear to have been variably up-dated probably due to argon loss caused by Cretaceous–Tertiary intrusives. Thermal and hydrothermal effects of these intrusions have probably reset the magnetization in the youngest formation of the volcanic pile. A composite palaeomagnetic pole calculated from the 30 site poles of the three lower formations (209° E, 81° S, $A_{95} = 4\frac{1}{2}°$), is in good agreement with mid to Late Cretaceous poles derived from rock units of the stable platform of South America. The use of Andean–Caribbean palaeomagnetic data however, to resolve small time-dependent polar shifts within the Cretaceous and thus to estimate the time of opening of the south Atlantic is questioned. Many of the Andean–Caribbean Cretaceous poles appear to have been affected by local tectonic rotation.

Geology of the Cretaceous units

The rocks immediately to the east of La Serena, north central Chile, provide an opportunity to study the palaeomagnetic history of a substantial portion of Cretaceous time for the Andean region of South America. The stratigraphy and structure of this region has been worked out in considerable detail by Aguirre & Egert (1965, 1970). The geological summary which follows is drawn from their studies, and supplemented by our own petrographic studies.

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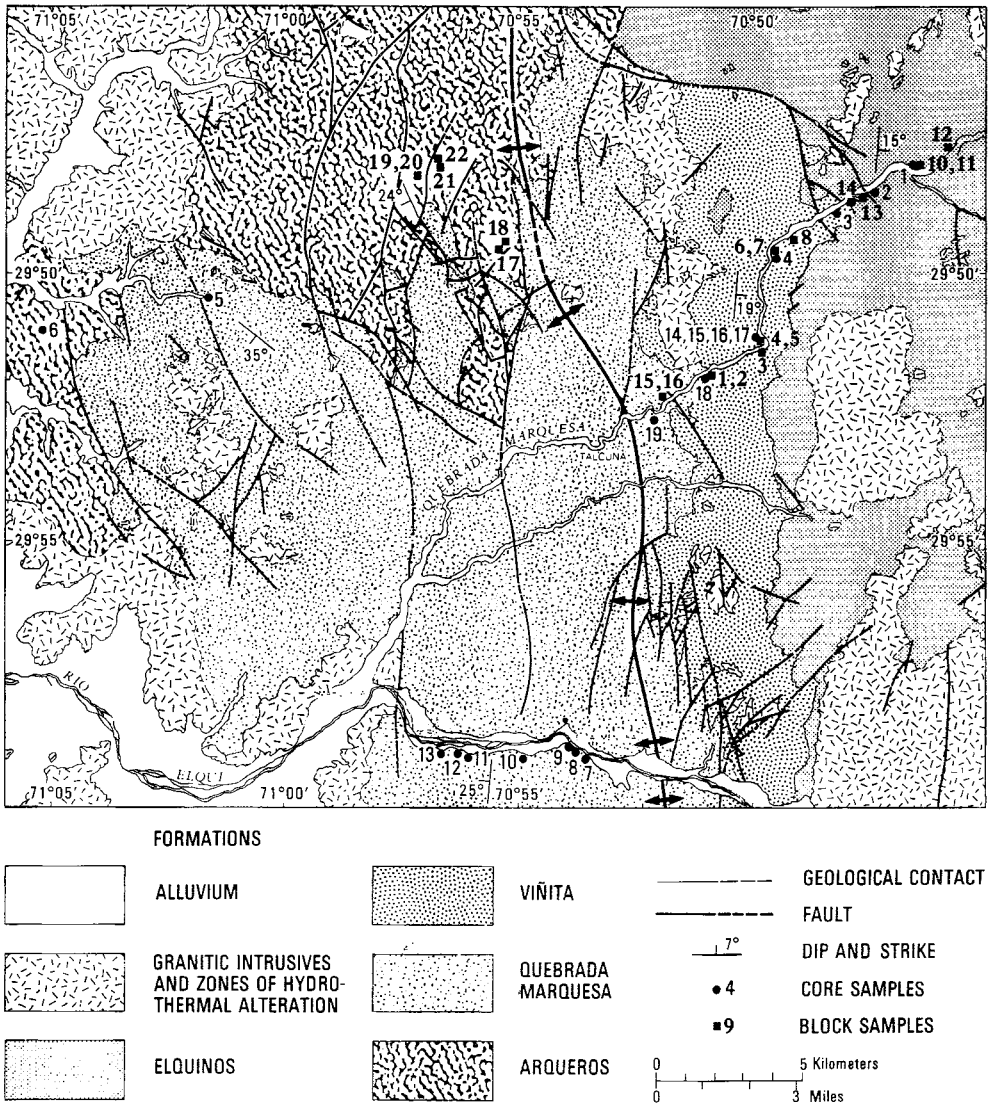


Figure 1. Geological map of the Quebrada Marquesa Quadrangle and the eastern part of the Lambert Quadrangle simplified from the maps of Aguirre & Egert (1965, 1970). The location of this area is indicated as site 12 on Fig. 7(b).

A simplified geological map (Fig. 1) shows about 6000 m of Cretaceous section, divided into four formations. The lower three formations have been further sub-divided into additional members. All four formations are composed primarily of lavas, tuff-breccias and tuffs of andesitic composition and volcaniclastic sandstones and shales. The units which we sampled are from the stratigraphically lowest to highest: the Arqueros Formation, the Quebrada Marquesa Formation, the Viñita Formation, and the Elquinos Formation.

The Arqueros Formation consists of three primarily volcanic members dominated by feldspar–phyric andesites separated by two primarily sedimentary members containing a shallow water marine fauna. That fauna places the formation in the Hauterivian–Barremian stages of the Early Cretaceous. Our sampling sites in this formation are in members 1, 3 and

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5 of Aguirre & Egert (1965). The base of the Arqueros Formation is not exposed, and therefore the foundation on which this volcanic pile rests is not known.

The Quebrada Marquesa Formation rests conformably on the Arqueros Formation and has been divided into four members. The lower three members contain a high proportion of tuff-breccias and tuffs of andesitic composition. Most of these volcanic products have undergone extensive high-temperature oxidation and the rocks exhibit colour tones in reds and purples. The third member in addition contains red coloured sandstones and conglomerates. The uppermost member consists dominantly of grey coloured basalt or andesite flows in which original phenocrystic olivine now pseudomorphed by earthy haematite gives the rocks a distinctive speckled appearance. The lowest member of the Quebrada Marquesa Formation contains a few tens of metres of shallow marine limestones and calcareous sandstones bearing a number of pelecypod forms which suggest a late Neocomian age for the formation. Sedimentary units higher in the formation contain plant remains, exhibit dessication cracks and rain-drop imprints and are therefore continental and are entirely lacking in fossils of stratigraphic value.

The overlying Viñita Formation, although structurally conformable with the Quebrada Marquesa Formation, bears a regional unconformable relationship to the older formation (Aguirre & Thomas 1964). The lowest member of the Viñita Formation is a conglomerate containing clasts derived from units in the Arqueros and Quebrada Marquesa Formations. The upper two members, both of which were sampled in the present study, contain grey coloured andesitic lavas and tuffs of grey and red colour. The Viñita Formation is largely, if not entirely non-marine. Poorly preserved indeterminate, fresh-water gastropods are the only fossils recovered from this formation; therefore its age is in doubt. Aguirre & Egert (1965) place it tentatively in the range late-Albian–early Coniacian.

The Elquinos Formation, entirely non-marine and also lacking in fossils, is composed of lavas and tuffs ranging from basalt to rhyolite. The rhyolite is an ash-flow occurring approximately 1400 m above the base of the formation and is exposed outside the limits of Fig. 1. All of our sampling sites are between the base of the formation and this distinctive ignimbrite unit. The ash-flow has served as a tool in making correlations of the Elquinos Formation with other formations of central Chile but these as well are not well defined in age. Some authors have suggested that the correlatives of the Elquinos Formation may be as young as Early Tertiary.

The major structure of the area is a broad anticline, broken by numerous high angle faults of N–S trend. Because of these faults, the stratigraphically lowest unit, the Arqueros Formation, shows only westerly dips. The Quebrada Marquesa, with both east- and west-limb exposures in the anticline, allows a fold test. The highest formations, the Viñita and Elquinos, lie on the east limb and show only homoclinal easterly dips. Both eastward and westward of the anticline, the Cretaceous units are invaded by widespread stocks and batholiths of diorite and granodiorite. Thus in spite of the great stratigraphic thickness and structural variety in this area, a fold test is only possible for the Quebrada Marquesa Formation.

The lavas and tuffs of all four formations are of andesitic composition in which plagioclase is the most abundant mineral both as a phenocryst phase and in the groundmass. In none of the rocks examined is the plagioclase completely unaltered; the minerals which partially replace the plagioclase are calcite, sericite, chlorite and epidote. The dominant ferromagnesian mineral is clinopyroxene. Primary hornblende was noted in only one of the 40 thin sections examined. Olivine is a minor porphyritic phase in some of the lavas but is largely replaced by serpentine–chlorite–haematite assemblages or serpentine–calcite–haematite assemblages. The primary ground-mass of most of the rocks examined consists of

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plagioclase, pyroxene and oxides but the secondary minerals calcite, chlorite and epidote are widespread. A complete range in oxidation classes of original titanomagnetite (Wilson & Haggerty 1966; Ade-Hall *et al.* 1968; Haggerty 1976) is present in these rocks with many of the grains in the tuffaceous rocks being of the high oxidation classes of 5 and 6. In oxidized grains of class 2 and class 3 the 'exolved' ilmenite lamellae are now wholly replaced by black sphene, a feature characteristic of burial metamorphism from the base of the laumontite zone to the epidote zone (Ade-Hall, Palmer & Hubbard 1971). All four formations have been reheated to conditions of approximately the prehnite–pumpellyite facies of metamorphism. This reheating was probably not a burial-type metamorphism inasmuch as we noted no correlation of secondary mineral assemblages with stratigraphy. Large masses of granodiorite intrude the Cretaceous rocks both to the east and west (Fig. 1). Small plugs of intrusive rocks, too small to illustrate on Fig. 1, (see Aguirre & Egert 1965, 1970) penetrate the volcanic formations of the map area. These volcanic formations may be rafted on an underlying granitic mass, in which case secondary minerals result from a diffuse contact metamorphism.

Palaeomagnetic sampling and laboratory methods

A collection of 22 block samples (LS series) was made in the map area in 1967 as part of a reconnaissance sampling of various Mesozoic units of northern Chile. In 1974 a second collection was obtained (QM series) using a portable rock drill in which generally four independently oriented cores (samples) were obtained at each of 19 additional sites. The azimuthal component of orientation of the field drilled samples was determined with a sun compass and checked with a Brunton compass corrected for the local published magnetic declination. Agreement of the sun and magnetic bearings to within 3° gives assurance that there is no gross orientation problem with the older collection (LS) series which was oriented by the magnetic method only.

In the laboratory, two to four specimens were prepared from each block sample and the field-drilled samples were cut into specimens with each core typically yielding two specimens. A polished thin section was made from each block sample and from one of the samples from each of the QM series sites. Small pairs of chips with a combined mass of about 0.5 g were prepared from selected samples for use in a Curie balance.

The early stages of the palaeomagnetic measurements were performed using a PAR spinner magnetometer. All measurements of remanence subsequent to the 30 mT af demagnetization step and all measurements of NRM following thermal or chemical demagnetization treatments were performed with a Schonstedt DSM-1 spinner magnetometer. Alternating field demagnetization was carried out to a maximum of 110 mT (peak) using apparatus similar in design to that described by Roy, Reynolds & Sander (1973). Two to four specimens from each block sample and one specimen from each of the 78 QM series samples were progressively demagnetized in steps to a routine maximum of 80 mT. Minimum dispersion within the range of a defined stable end-point was used as the criterion for selection of the cleaned direction to represent each site. Step-wise thermal demagnetization was carried out on selected samples and sites. In some experiments the thermal demagnetization was performed on specimens which had previously been subjected to the partial alternating field treatment; in other experiments the thermal demagnetization was done on specimens which had not been so treated. The thermal demagnetization was done in furnaces employing nested mumetal shielding. One of the demagnetizers is a Schonstedt TSD-1 model; the other is a hand-crafted unit in which the specimens can be slowly rotated along one axis during the cooling cycle. This rotation minimizes the possible acquisition of

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Table 1. Palaeomagnetic directional data – Arqueros Formation.

Site/ sample	N	af/T	R	k	D (°)	I (°)	D Structurally (°)	I corrected (°)	Pole position Long.	Lat.	dp (°)	dm (°)
LS 17	3	10 mT	2.975	81.0	014½	−54	024	−51½	148½° W	69° S		
LS 18	3	80 mT	2.994	354.0	000½	−28½	004½	−28	126° E	74½° S		
LS 19*	3	60 mT	2.994	336.0	114½	+ 66½	181½	+ 72				
LS 20	3	50 mT	2.991	211.0	337	−43½	358	−39	96° E	82° S		
LS 21	3	20 mT	2.999	3578.0	352	−37	003	−27½				
LS 21	3	625°C	2.995	397.0	347	−42	001	−33½	114½° E	78½° S		
LS 22	3	50 mT	2.998	1001.0	005½	−55½	022	−41½	173° W	69½° S		
QM 6	5	60 mT	4.985	262.0	014	−47½	001	−53	84° W	86° S		
QM 6	5	600°C	4.982	220.0	011	−49	357½	−54				
Mean	6		5.844	32.0	358½	−46						
Mean	6		5.874	40.0			008	−41½	161° E	80½° S	8.4	13.2
Mean	6		5.877	41.0					165° E	81° S	A_{95} = 10.6°	

* The value of k for this sample is less than 10 for all demagnetization steps other than the 60 mT treatment and is not used in the calculation of the formational mean direction and mean pole.

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Table 2. Palaeomagnetic directional data – Quebrada Marquesa Formation.

Site/ sample	N	af/T/hr	R	k	D (°)	I (°)	D Structurally (°)	I corrected (°)	Pole position Long.	Lat.	dp	dm
LS 1	2	60 mT	1.999	2504.0	038½	−44	024½	−47	159½° W	68½° S		
LS 1	2	600°C	1.996	230.0	044½	−44½	030½	−48½				
LS 2	3	60 mT	2.997	717.0	014	−41	002½	−38	124° E	81½° S		
LS 15	3	70 mT	2.999	6780.0	027	−41½	008	−47½	168½° W	83° S		
LS 15	2	116 hr	1.996	285.0	021½	−38½	004½	−43½				
LS 16	3	30 mT	2.999	1752.0	044½	−51	018½	−61½				
LS 16	3	550°C	2.991	212.0	038	−50	012½	−58½	115° W	76° S		
LS 16	2	116 hr	1.998	505.0	036	−55½	004	−62½				
QM 5	5	10 mT	4.918	49.0	004½	−58½	017½	−28	160½° E	68° S		
QM 8	4	20 mT	3.934	45.0	351	−46	017½	−46½	165½° W	74½° S		
QM 10	4	30 mT	3.959	74.0	334½	−41½	359	−49½	001° W	89° S		
QM 10	4	600°C	3.963	82.0	331½	−40	355	−49½				
QM 12	4	15 mT	3.990	295.0	345	−47	013½	−49½	156° W	78° S		
QM 13	4	70 mT	3.993	421.0	357	−47½	024	−45	164° W	69° S		
QM 13	4	600°C	3.986	207.0	356	−46½	022	−44½				
QM 18	4	60 mT	3.988	254.0	032½	−32½	019	−38	177½° E	71° S		
QM 19	3	20 mT	2.999	4371.0	020½	−54½	020½	−54½	139° W	72° S		
QM 19	3	600°C	2.996	516.0	023	−56½	023	−56½				
Mean	11		10.559	23.0	011½	−48						
Mean	11		10.842	63.0			014½	−46	168½° W	77° S	4.9	7.5
Mean	11		10.849	66.0					166½° W	77° S	A_{95} = 5.6°	

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Table 3. Palaeomagnetic directional data – Vĩñita Formation.

Site/ sample	N	af/T	R	k	D (°)	I (°)	D Structurally corrected (°)	I corrected (°)	Pole position Long.	Lat.	dp (°)	dm (°)
LS 3	3 *	20 mT	2.995	406.0	031	−56	358	−61½	065° W	77° S		
LS 3	3	600°C	2.989	181.0	028½	−57	355	−61				
LS 4	3	60 mT	2.987	159.0	023	−41½	004	−46	171½° E	86° S		
LS 5	2	40 mT	1.999		044½	−48	021	−59				
LS 5	2	625°C	1.999	1108.0	030	−45½	008	−52	135° W	82½° S		
LS 6	1	40 mT	–	–	029½	−40½	011	−47½	166° W	80½° S		
LS 7	3	70 mT	2.994	347.0	021½	−45½	359½	−49½	038½° W	89½° S		
LS 8*	3	80 mT	2.969	65.0	040½	−48½	020	−61	118½° W	70° S		
LS 14*	3	80 mT	2.999	3443.0	007½	−53	346	−48½	016½° E	77½° S		
QM 3	4	40 mT	3.965	87.0	004	−66½	341½	−72	050½° W	60° S		
QM 3*	4	400°C	3.958	72.0	009½	−66½	349	−72½				
QM 4	4	70 mT	3.947	56.0	012	−52	009½	−55½	121° W	80° S		
QM 14	4	80 mT	3.994	516.0	041	−46	020½	−54½	138½° W	72° S		
QM 15	4	30 mT	3.967	90.0	011½	−52½	347½	−50½	009° E	79° S		
QM 16	4	60 mT	3.987	150.0	043½	−42	025½	−51½	148° W	68° S		
QM 17	4	50 mT	3.988	249.0	050½	−47½	029	−59	130½° W	64½° S		
Mean	13		12.750	48.0	027½	−50						
Mean	13		12.777	54.0			006½	−55½	111° W	82° S	6.0	8.1
Mean	13		12.623	32.0					108° W	81½° S	$A_{95}=7.5°$	

* Thermal demagnetization was attempted on these sites but the samples broke up in the 400 or 500°C run.

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Table 4. Palaeomagnetic directional data – Elquinos Formation.

Site/ sample	N	af/T	R	k	D (°)	I (°)	D Structurally (°)	I corrected (°)	Pole position Long.	Lat.	dp (°)	dm (°)
LS 10	3	20 mT	2.996	488.0	008	−49	348½	−36½	058° E	76° S		
LS 11	3	30 mT	2.991	215.0	011	−32½	358	−26	102° E	74° S		
LS 12	3	30 mT	2.997	670.0	226½	+ 77	158½	+ 76½				
LS 12	4	600°C	3.829	18.0	192	+ 59½	169	+ 55½	017½° W	79° S		
LS 13	3	50 mT	2.997	591.0	009½	−66	338	−61				
LS 13	3	600°C	2.998	884.0	006½	−62½	339½	−57½	013° W	71° S		
QM 1	4	50 mT	3.987	238.0	016½	−46	006	−26	128½° E	73° S		
QM 2	4	80 mT	3.999	2601.0	010½	−51½	007½	−46½	177½° W	83° S		
Mean	6		5.908	55.0	011	−49½						
Mean	6		5.805	26.0			356	−41½	076° E	83° S	10.7	16.5
Mean	6		5.841	32.0					065° E	84° S	$A_{95} = 12.1°$	

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small PTRM's in one plane due to incomplete dc field cancellation. Temperature calibration experiments, temperature gradient measurements and duplicate runs in this furnace and in the TSD-1 indicate that temperatures (Tables 1–4 and subsequent figures) are conservatively estimated to be accurate to $\pm 5^{\circ}$C. Chemical demagnetization involving successive immersions in 10 N HCl, was performed on specimens of two samples of red sandstones from the Quebrada Marquesa Formation. Preferential destruction of the red pigment, ahead in time of the larger detrital haematite grains, was aided by specimen serration with a small thin-walled drill in the manner described and illustrated by Morris & Carmichael (1978). J_{sat} versus T curves were obtained by heating and cooling small chips in air in a constant gradient field by a null method in a horizontal translation balance; at the null position the applied field used was 0.8 T.

Palaeomagnetic results

For this collection 77 per cent of the samples have median destructive af fields of NRM in excess of 30 mT and 33 per cent of the total have MDF's in excess of 80 mT. In general, af demagnetization serves to increase the precision of each formational mean direction but without producing large directional changes in any individual sample or site mean i.e. the directions are well grouped initially.

The samples from three field-drilled sites in the Quebrada Marquesa Formation (QM 7, 9 and 11) and the specimens from one block sample from the Arqueros Formation (LS-19) proved to be either magnetically unstable or showed poor within-site directional agreement throughout the af demagnetization treatment. These samples and sites are rejected from all subsequent analysis.

The palaeomagnetic directional results are given in Tables 1–4 and are graphically displayed in Fig. 2. In the statistical tables where two or more entries appear for a sample or site corresponding to results obtained by af, thermal or chemical treatment, the entry having a calculated pole is the result used in the graphical display of Fig. 2.

With respect to Fig. 2 and Tables 1–4, the following observations can be made:

(1) All samples and sites, with the exception of LS-12 which is the stratigraphically youngest sample in the youngest formation, are of normal polarity.

(2) Although the NRM results from the Arqueros Formation are better grouped than are the 'cleaned' directions, the NRM results are degraded by the application of the structural tilt correction whereas the 'cleaned' directions are slightly better grouped when the bedding correction is applied.

(3) Application of the structural tilt correction to the 'cleaned' Quebrada Marquesa data increases the precision of the mean direction significantly (significant at the 95 per cent confidence level, McElhinny 1964). This indicates that a Cretaceous magnetization has been isolated. The precision of the mean from the Arqueros and Viñita Formations is also increased upon application of the structural correction but not significantly. A significant improvement is not to be expected however, as the latter formations have homoclinal dips.

(4) Application of the structural tilt correction to the sites within the youngest formation, the Elquinos Formation, results in a degradation of the precision of the mean direction. This is probably the result of several of the sites having been remagnetized by post-Elquinos intrusions which are particularly abundant in the eastern part of the area of study.

It has been established for some time that the reversal frequency during the Cretaceous period differed markedly from that of the Tertiary and that Cretaceous time was dominated by fields of normal polarity (Sasajima & Shimada 1966; Helsley & Steiner 1969). A 25 Myr

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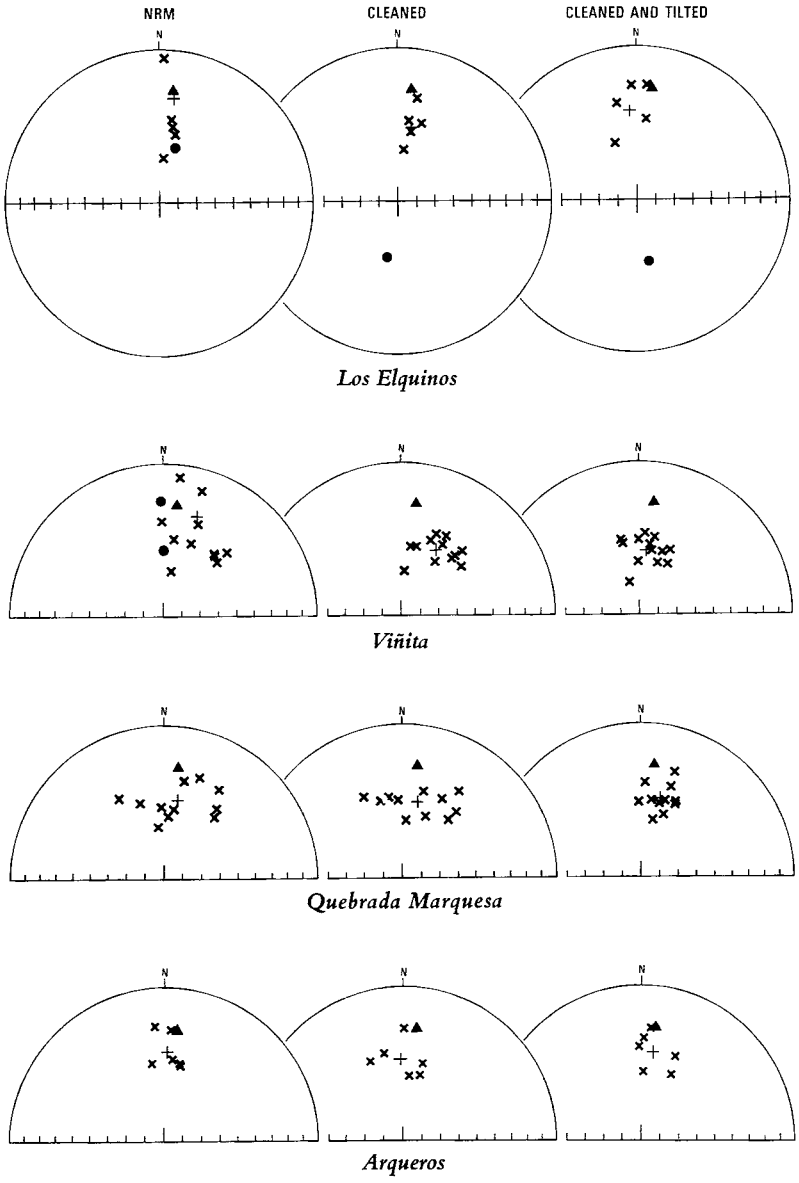


Figure 2. Equal area nets showing the directions of NRM for all acceptable samples/sites from each of the formations studied. In the middle column are shown the same sample/site directions after af or thermal demagnetization. In the third column these cleaned directions are shown with the structural tilt correction applied. A fold test, with positive results, was only possible for the Quebrada Marquesa Formation. North-seeking directions are plotted as X's (dots) on the upper (lower) hemisphere. The solid triangle is the present Earth's field direction at La Serena.

uninterrupted normal polarity zone extends from the Aptian to the Coniacian stages of the Cretaceous as inferred from seafloor magnetic anomaly patterns (Larson & Pitman 1972). This time period of normal polarity is compatible with the known age of the Quebrada Marquesa and Viñita Formations. However, one might possibly expect to find some reversely magnetized sites in the Arqueros Formation if its age assignment to the Neocomian is correct. All of our samples from the Arqueros Formation are of normal

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polarity which may indicate that the magnetizations which we have isolated are primary and were acquired in normal polarity fields during this time of normal polarity bias (Irving & Pullaiah 1976) or that some or all of them were originally reversely magnetized but have subsequently been remagnetized in later Cretaceous normal polarity fields. Therefore detailed progressive thermal demagnetization was performed on a substantial part of the collection in an effort to uncover any reversely magnetized components which might be hidden in the total NRM. Progressive chemical leaching experiments were also performed on several suitable red sandstones for the same purpose. A few typical results of these experiments are given below.

In Fig. 3 are shown the results from af and thermal demagnetization on two specimens from the Quebrada Marquesa sample LS-1. In terms of remanence coercivity LS 1-1 is very typical of much of the collection with only small directional changes experienced during removal of the low coercivity components; substantial remanence exceeds the 80 mT limit of the experiment but the thermal treatment (LS 1-4) reveals no other distinct directional components which can be separated on the basis of blocking temperatures.

In Fig. 4 are shown the results of sequential af and thermal demagnetization on specimens from the Arqueros and Quebrada Marquesa Formations. Judging from the low values of saturation magnetization and the hard coercivity of NRM, haematite is the sole magnetic phase in these rocks. Both exhibit square shouldered unblocking curves with no evidence of anything more than a single component of remanence.

Chemical demagnetization could be performed on only a small fraction of the collection as not much emphasis was placed on the sampling of interbedded sedimentary rocks. Member 3 of the Quebrada Marquesa Formation is composed almost entirely of sedimentary units, however, and two samples of red sandstones were collected from it. Specimens of these samples, LS-15 and LS-16, were subjected to repeated immersions in 10 N HCl followed by washing and drying in field-free space. Because these rocks contain calcite, as a cement and as a replacement of detrital grains, destruction of the red pigment throughout

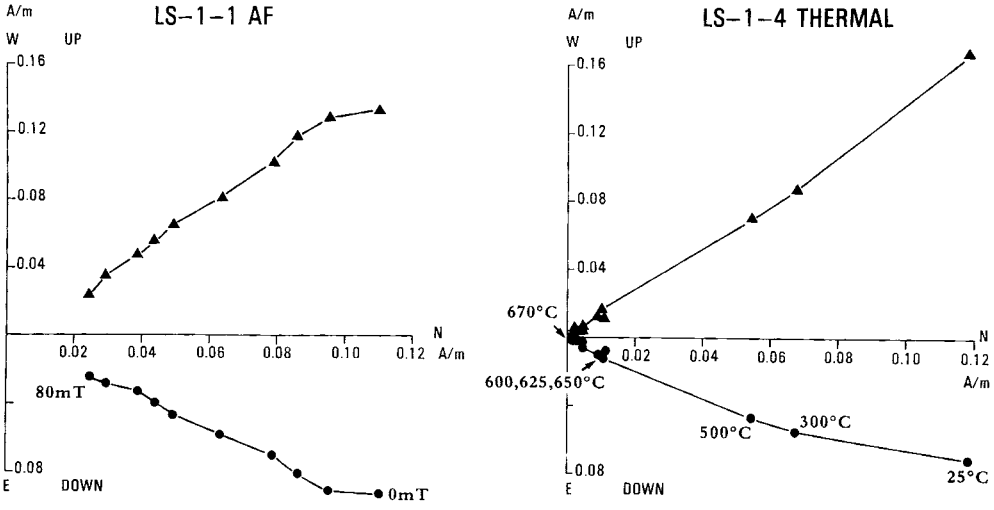


Figure 3. Orthogonal demagnetization projection (Zijderveld 1967) of the successive endpoints of the magnetization vector in two normally magnetized specimens from the Quebrada Marquesa Formation. LS-1-1 was partially demagnetized by the af method whereas in LS-1-4 the thermal demagnetization method was used. Solid dots represent directions on the horizontal plane whereas triangles represent projections on the north–south vertical plane. Bedding correction not applied.

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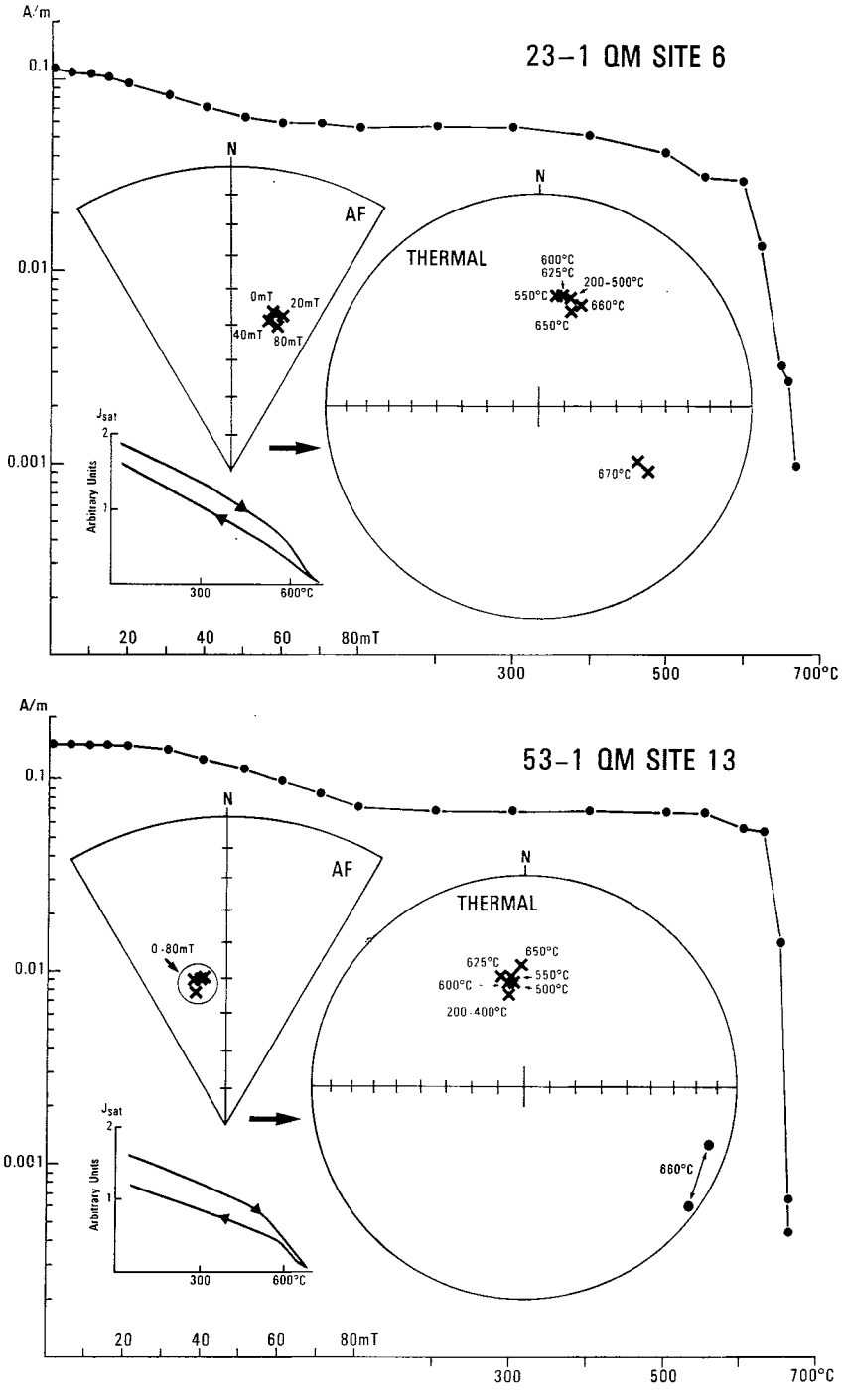


Figure 4. Progressive demagnetization of normally magnetized specimens from the Arqueros Formation (23-1) and from the Quebrada Marquesa Formation (53-1) in which thermal demagnetization follows af demagnetization. Normal directions are represented as X's in the equal area plots; north-seeking down directions are represented by solid dots. In this and in subsequent stereonet plots, the structural tilt correction has not been applied. J_{sat} versus T curves constructed from small chips of these samples are shown as small insets in the figures.

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the specimens was relatively rapid. The results from the chemical leaching of two serrated specimens from each of the samples are shown in Fig. 5. Within specimen homogeneity remained high up to the total of 164 hr immersion, followed by greater within-specimen inhomogeneity and between-specimen dispersion at longer immersion times. Thus not much significance should be attached to the slight directional shifts in both samples following the 340 hr treatment. In general the limited chemical demagnetization work confirms the end points achieved in these samples by either the af or thermal demagnetization method (Fig. 5 and Table 2).

Only two samples exhibited evidence which might suggest the remanence in these samples consists of normal and reversed components (Fig. 6). LS-21 is from the Arqueros Formation and LS-5 is from the Viñita Formation. Considering LS-21-4 first, af demagnetization to 80 mT reduces the NRM intensity to 8 per cent of its initial value without significant directional change. However, subsequent thermal treatment results in a slight intensity rise

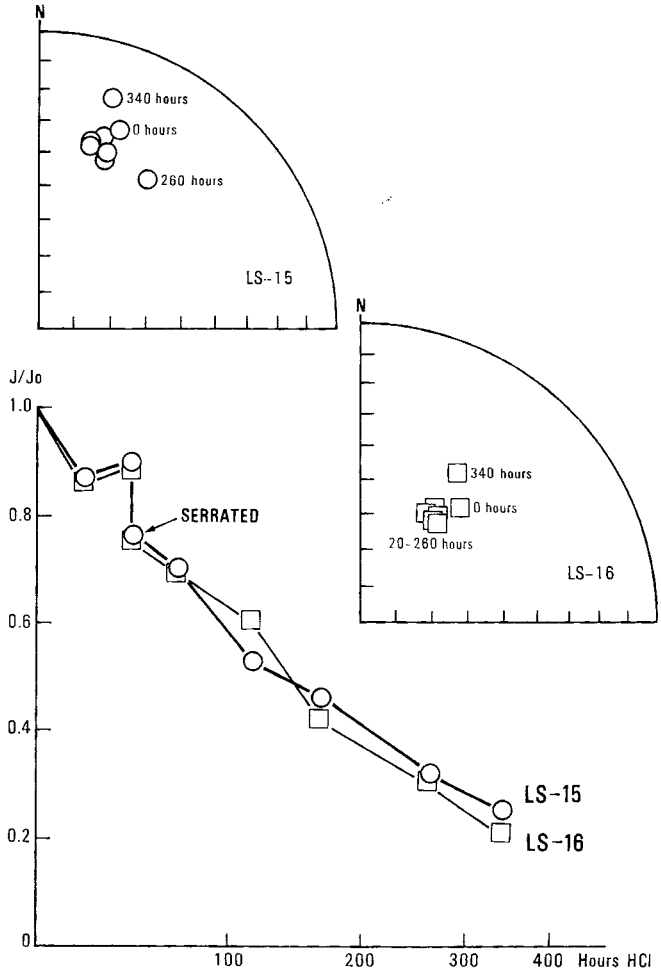


Figure 5. Average directions and relative intensities of two specimens from LS-15 (circles) and LS-16 (squares) are shown after successive immersions in 10 N HCl for the cumulative times indicated on the horizontal axis of the demagnetization graph. All directions on the portions of the equal area nets are on the upper hemisphere.

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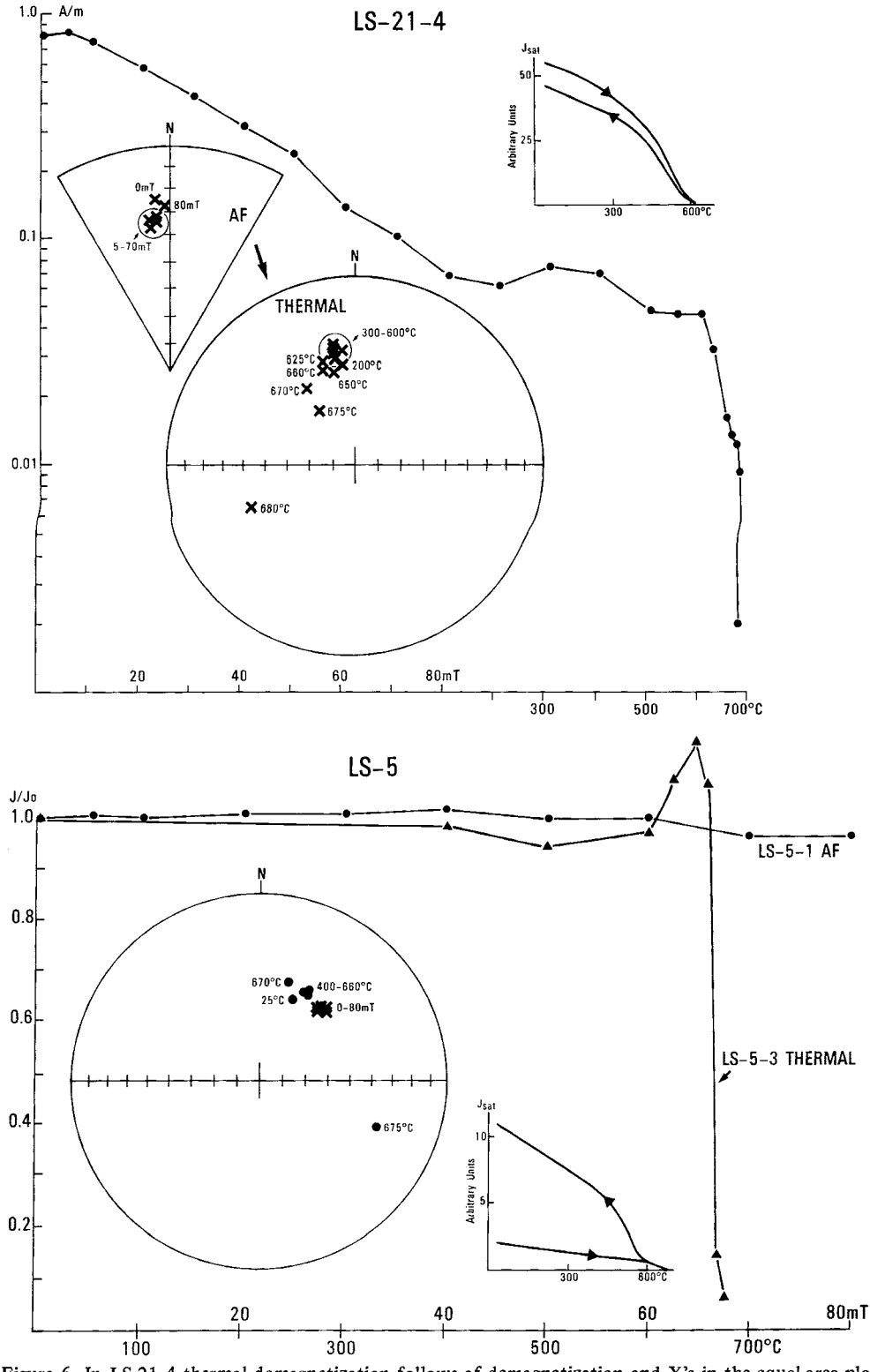


Figure 6. In LS-21-4 thermal demagnetization follows af demagnetization and X's in the equal area plot represent north-seeking directions on the upper hemisphere. In the equal area plot for LS-5, X's refer to LS-5-1 and dots refer to LS-5-3; both symbols represent directions on the upper hemisphere.

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without directional change over the 300–400°C interval. This suggests that an antiparallel component is being removed over this temperature range. The lower part of Fig. 6 shows the results obtained from two specimens of LS-5 one of which was examined by the af method and the other by the thermal demagnetization technique. The af method effects no directional or intensity change in LS-5-1. Although LS-5-3 undergoes no significant directional changes until it is effectively unblocked at the 670 and 675°C runs, there is a pronounced intensity rise in the 600–650°C interval. This might suggest again, as with LS-21-4, removal of an antiparallel component, i.e. a reversed component over this temperature range. However, there is another plausible explanation. The J_{sat} versus T curve (inset of Fig. 6) exhibits a pronounced increase in saturation magnetization upon cooling from high temperature. This indicates that the heating process itself results in the production of a strongly magnetic phase which is not present prior to the heating. Thus the evidence of reversed components is meagre. The simplest interpretation of all the data is that the characteristic normal magnetizations are primary but in a few samples these normal magnetizations may have been overprinted by a small secondary reversed component. It is also of course possible that the characteristic normal magnetizations are secondary and that only in a few samples has the characteristic primary reversed magnetization survived. In either case, however, the secondary component is antiparallel to the primary component and therefore the age difference between them is likely to be small; the axis of magnetization is thus likely to be a Cretaceous one.

K–Ar dating

Thin section examination was used in the initial selection of samples for K–Ar analysis. No samples could be found in the collection which were completely free of secondary mineral alteration. In fact, all our samples from the Arqueros formation have such a heavy sericite overprint on the plagioclase that no representatives from this formation were selected for analysis. Initially only three samples were selected, one from each of the three younger formations. The calculated ages from these three samples were all younger than

Table 5. K–Ar analytical data listed in stratigraphic order.

Sample	Formation	K (per cent)	$^{40}Ar_{rad}$ ($\times 10^{-6}$ cm^3 g^{-1})	Age (Myr)
LS-11	Elquinos	0.936	2.187	59.1
LS-14	Viñita	2.050	5.250	64.7
LS-2	Quebrada Marquesa	1.535	3.788	62.4
LS-1	Quebrada Marquesa	2.104	5.502	66.0
19 QM site 5	Quebrada Marquesa	0.663	2.431	91.9
54 QM site 13	Quebrada Marquesa	1.837	4.454	61.3
31 QM site 8	Quebrada Marquesa	0.913	2.965	81.7

Notes:

The constants used were: $\lambda_e = 0.581 \times 10^{-10}$ yr^{-1}; $\lambda_\beta = 4.962 \times 10^{-10}$ yr^{-1}; atomic $^{40}K/K = 1.167 \times 10^{-4}$.

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expected. Accordingly, additional samples were selected from a single formation, the Quebrada Marquesa Formation, to test whether argon loss or the presence of initial argon was an important factor in these samples.

All samples studied were crushed to pass 20 mesh with powders finer than 100 mesh being discarded. Each sample was then split into two portions, one of which was further crushed to pass 100 mesh and used for potassium analysis. The other portion was loaded into an argon extraction system immediately after crushing, and baked for 18 hr at 190°C, followed by an argon extraction.

The results are given in Table 5. The older apparent ages appear in the oldest formation, the Quebrada Marquesa. However, younger ages equal to those in the younger Viñita and Elquinos Formations are present in samples from the Quebrada Marquesa Formation. Because a significant unconformity is present between the Quebrada Marquesa and the Viñita Formation and because the apparent ages within the Quebrada Marquesa Formation are not stratigraphically ordered, none of the ages given in Table 5 is considered to give a valid estimate of the time of formation. An isochron plot of the Quebrada Marquesa data shows considerable scatter of the data points with no evidence that the samples share a common 'initial' argon component. The discordance is most readily explained by variable argon loss subsequent to deposition.

Discussion

Individual formation poles are given in Tables 1–4. Because the data from the Elquinos Formation are degraded by the application of the structural tilt corrections, it is inferred that some or all of the sites within this formation have acquired post-Elquinos magnetizations. The data from this formation are rejected.

The Quebrada Marquesa pole (Table 2) and the Viñita Formation pole (Table 3) are different at the 95 per cent confidence limit. This may reflect polar wander, incomplete averaging of secular variation or some combination of secondary overprints and local tectonic complexities. Inasmuch as these possible causes cannot be resolved, a composite Cretaceous palaeomagnetic pole is calculated (209° E, 81° S, $A_{95} = 4½°$) by averaging the 30 valid site poles from the Arqueros, Quebrada Marquesa and Viñita formations. This pole is very close to the composite pole (220° E, 85° S, $A_{95} = 6°$) recently reported by Valencio *et al.* (1977) for the time interval: late Early Cretaceous–Late Cretaceous. However, the stratigraphic age of the Arqueros Formation (Hauterivian–Barremian) is included in their definition of their early Early Cretaceous 'age group' of poles. The Arqueros pole does not fall in this 'age group' but is indistinguishable from those placed in the younger 'time group' – late Early Cretaceous to Late Cretaceous. Two explanations can be offered: remagnetization and tectonic rotation, as described below.

Late Cretaceous remagnetization of the Arqueros Formation

In spite of careful laboratory work directed at uncovering secondary components in the La Serena area rocks, this possibility of remagnetization of the Arqueros Formation in a later Cretaceous normal polarity field, still exists. The distinctive porphyritic andesites of the Arqueros Formation contributed clasts to the lowest conglomerate-rich member of the Viñita Formation. Therefore future palaeomagnetic work in this area could profitably be directed to performing a conglomerate test using these distinctive clasts. This test would establish whether or not the andesites of the Arqueros Formation have retained a pre-Late Cretaceous magnetization.

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Local tectonic rotation in orogenic belts

In Table 6 we tabulate South American Cretaceous palaeomagnetic poles. Rather than subdividing this list on the basis of age within the Cretaceous period we have grouped the poles on the basis of tectonic setting of site locations: A, sites overlying the Precambrian stable platform and B, sites from the Andean–Caribbean fold belts. These sites, together with their corresponding poles are graphically displayed in Fig. 7(a) and (b). Although some sites from the orogenic belt yield poles in agreement with those from the stable platform, many are discordant and a few are very discordant. We interpret this to mean that sites within fold

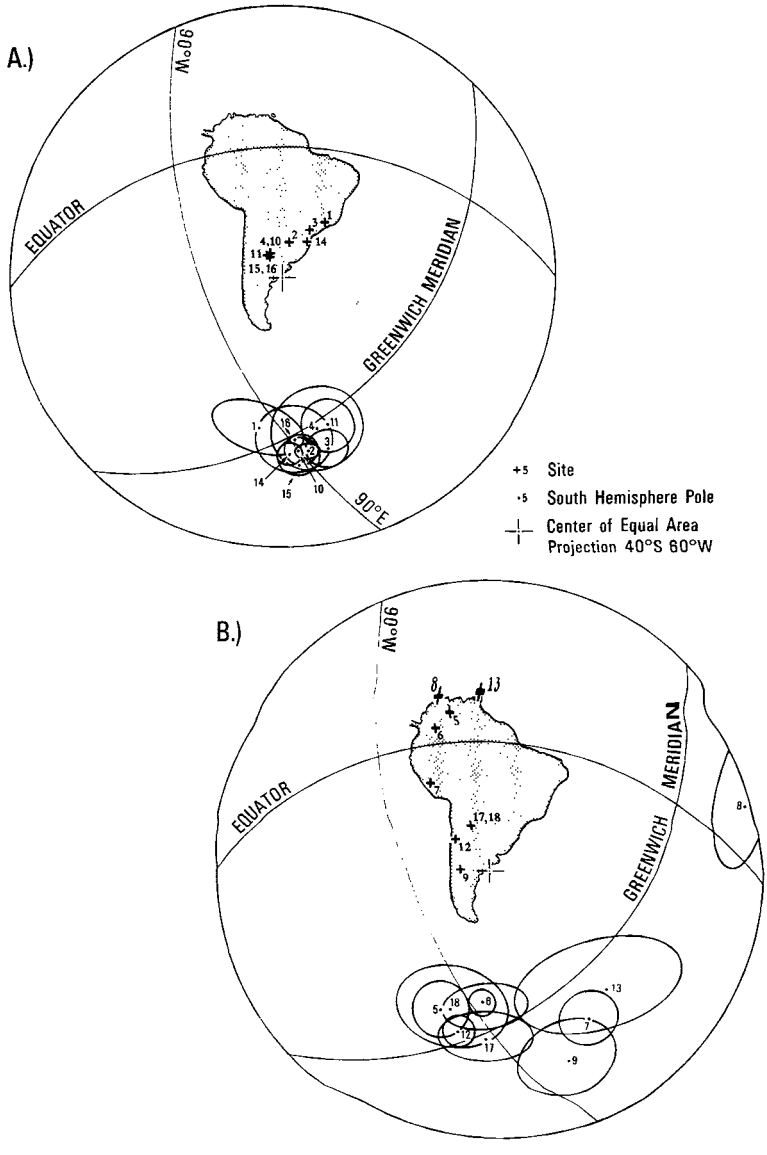


Figure 7. Cretaceous sites and their corresponding poles numbered as in Table 6, (a) displays poles from sites overlying Precambrian basement whereas (b) displays poles from the Andean Caribbean orogenic belt. Pole 12 is that derived from this study. The poles from the orogenic regions are more widely dispersed than those from the stable platform.

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Table 6. Cretaceous palaeomagnetic poles for South America.

Pole no.	Rock unit	Locality Long. (°)	Lat. (°)	Age	Palaeomagnetic pole Lat. (°)	Long. (°)	*dp* (°)	*dm* (°)	A_{95} (°)	Reference
1	Pocos de Caldas volcanics	Brazil; 21.9 S,	046.6 W	Late Cretaceous	82 S	127 W	8	13		Opdyke & MacDonald (1973)
2	Serra Geral basalt	Uruguay; 29 S,	057 W	Early Cretaceous	83 S	068 E	2.0	3.0		Creer (1958)
3	Serra Geral basalt	Brazil; 25 S,	050 W	Early Cretaceous	78 S	054 E	4.0	6.7		Creer (1962a)
4	Cerro Colorado volcanics	Argentina; 32 S,	064 W	Early Cretaceous	81 S	014 E			13	Valencio (1972)
5†	Yeguera formation	Venezuela; 09.0 N,	068.1 W	Cretaceous	75 S	124 W			8★	Creer (1970)
6†	Aptraxa formation	Colombia; 05.5 N,	073.4 W	Cretaceous	80 S	072 W			4★	Creer (1970)
7†	Vinchos and Marocoche strata	Peru; 11 S,	076 W	Cretaceous	63 S	030 E			8★	Creer (1970)
8‡	La Teta formation	Colombia; 11.8 N,	071.9 W	Early Cretaceous	10 N	021 E	9.6	19.1		MacDonald & Opdyke (1972)
9	Huitrinian redbeds	Argentina; 39 S,	071 W	Late Cretaceous	66 S	062 E			13.5	Creer (1962b)
10	Cerro Rumipalla formation	Argentina; 32 S,	064 W	Cretaceous < 118 Myr	85 S	082 E			5	Cited in Valencio (1974)

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11	Rio de Los Molinos dykes	Argentina; 31.8 S, 064.5 W	Cretaceous (Jurassic to early Tertiary)	78 S	013 E			8	Linares & Valencio (1974, 1975)
12	Arqueros, Quebrada Marquesa, Viñita combined	Chile; 29.8 S, 070.9 W	Cretaceous	81 S	151 W			4.4	This study
13	Hawk's Bill formation	Tobago; 11.2 N, 060.8 W	Cretaceous > 113 Myr	57 S	016 E	13.7	27.4		MacDonald & Van Horn (1977)
14	Serra Geral basalt	Brazil; 28.4 S, 049.5 W 29.4 S, 050.3 W	Early Cretaceous (119 Myr)	85 S	115 E			3.7	Pacca & Hiodo (1976)
15	Rumipalla volcanics, S. Condores group	Argentina; 32.2 S, 064.1 W	Early Cretaceous	83 S	095 E			5.9	Recalculated from Vilas (1976)
16	Cerro Colorado volcanics, S. Condores group	Argentina; 32.2 S, 064.1 W	Early Cretaceous	88 S	034 E			10.9	Recalculated from Vilas (1976)
17	Las Curtiembres formation, Pirgua subgroup	Argentina; 25.75 S, 065.83 W	Late Cretaceous (77 Myr)	88 S	154 E	8	12		Valencio *et al.* (1977)
18	La Yesera formation Pirgua subgroup	Argentina; 25.75 S, 065.83 W	Early Cretaceous (98 and 114 Myr)	76 S	120 W	12	17		Valencio *et al.* (1977)

Notes:

* Estimated by doubling circular standard error.

† Natural remanent magnetization.

‡ Northern hemisphere pole; all other poles in this table are southern hemisphere poles, A_{95} is the half-angle of the cone of 95 per cent confidence about the mean of virtual geomagnetic poles.

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belts are subject to tectonic rotations – rotations which are not always evident with the tools of classical structural geology. Thus the use of palaeomagnetic data from orogenic belts to detect or confirm time-dependent polar shifts is likely to be hazardous.

In conclusion, we attribute the dispersion of Cretaceous poles for sites in the orogenic zones to local tectonic rotations. On the basis of evidence presently available from the tectonically stable platform of South America, no clear pattern of polar shift during the Cretaceous is perceptible. The best estimate of the mean Cretaceous pole for South America (mean of poles of Fig. 7a) is 85° S, 052° E, $A_{95} = 4\frac{1}{2}°$. Relative to that pole, the Serena Cretaceous volcanic section of central Chile appears to have undergone only a minor clockwise rotation, of approximately $15° \pm 11°$.

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